



UNIVERSITÀ DI PARMA

Applications of Machine Learning and spectral analysis to the study of natural metabolites

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What we will see

- How Machine Learning can solve problems in chemistry & biology
- Use of Neural Networks to obtain
 - **Molecular embeddings** : learning **co-occurrence patterns** of chemical structures
 - **Protein embeddings** : learning of **sequential patterns** of amino-acids that reflects the structure and the functional properties of the protein

What we will see

- How these embeddings can be used for
 - **Metabolite identification** : similarity between molecules
 - **Bioactivity prediction** : classification of molecule + protein interaction (Active/Inactive)
- **Application:**
 - NBFC - B4Web Project (2025)

B4Web project

- **Goal:** study of the **biodiversity** of Italian plants
 - **which molecules can be extracted**
 - classification of
 - plants: taxonomy
 - metabolites: type of molecule produced by a plant
 - metabolites bioactivities: effect of a metabolite on a living being
- **Create a Database** to store these informations

B4Web project

- **Research interest** in plant metabolites:
 - **Drug** development
 - **Natural supplements** and beauty products
- **Nagoya protocol** (2010)
 - **international agreement** under the Convention on Biological Diversity (CBD)
 - regulates **Access and Benefit-Sharing** (ABS) for **genetic resources**
 - **If you use** genetic resources (**plants**, animals, microbes, etc.) or traditional knowledge originating from a certain country, you must:
 - Get permission
 - Share benefits (no “biopiracy”)

B4Web project

- Partners
 - Università del Piemonte Orientale (coordinator)
 - Museo Civico di Verona → botanists - sampling plants
 - Università di Verona → Laboratory - metabolite identification
 - Università di Pisa → Laboratory - bioactivity identification
 - **Università di Parma** → **computational research**
 - Inferendo → Database backend - frontend

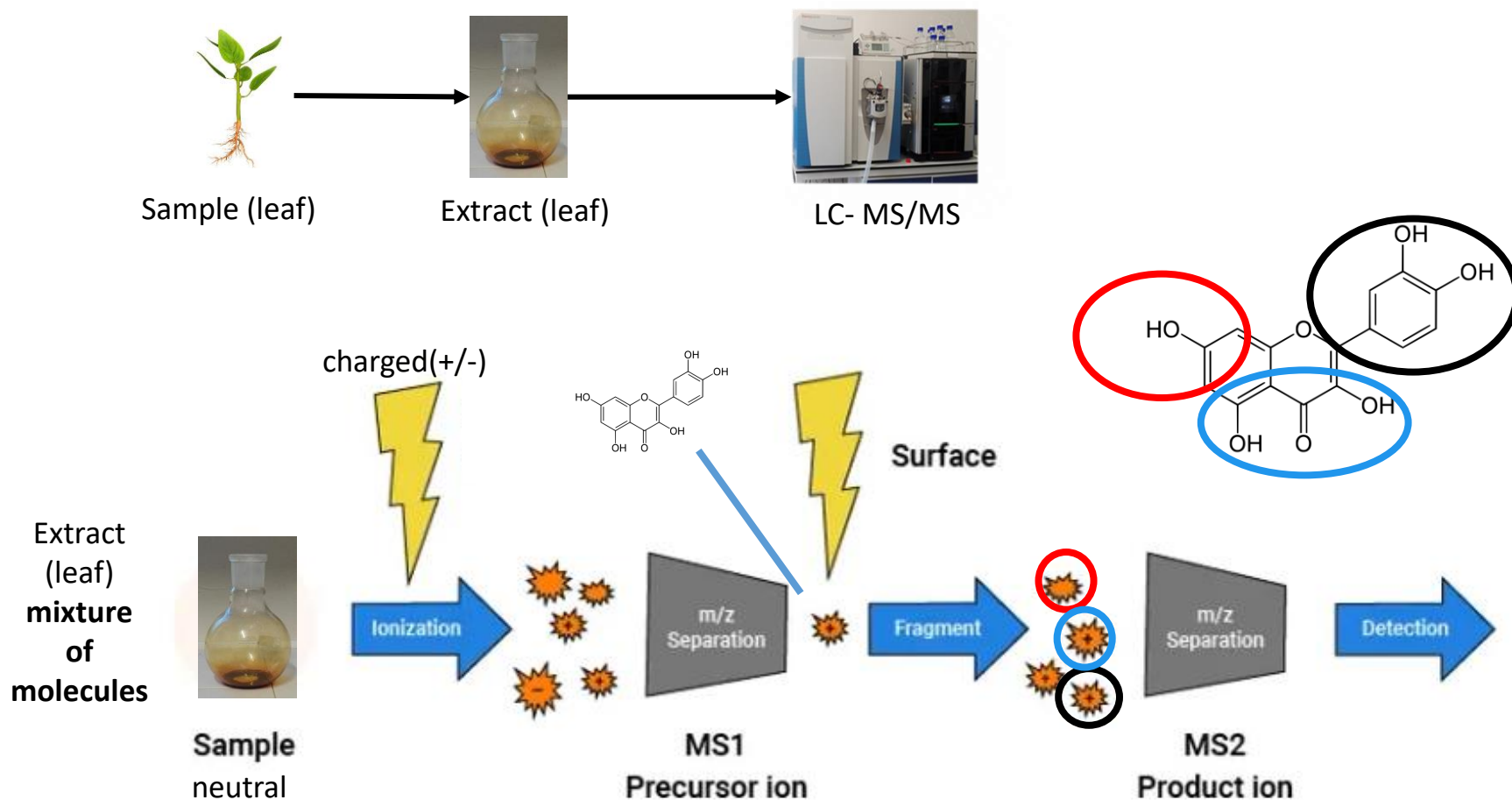
UniPR - Our research

- Guideline : “develop predictive algorithms”
- Research :
 - Develop a tool for the WebApp
 - **Use Machine Learning** for Metabolite Identification and Bioactivity Prediction
- Challenges:
 - Molecules data are **complex** → mass spectrometry profiles
 - Bioactivity (aka “Bioassay”) → very **vague definition**

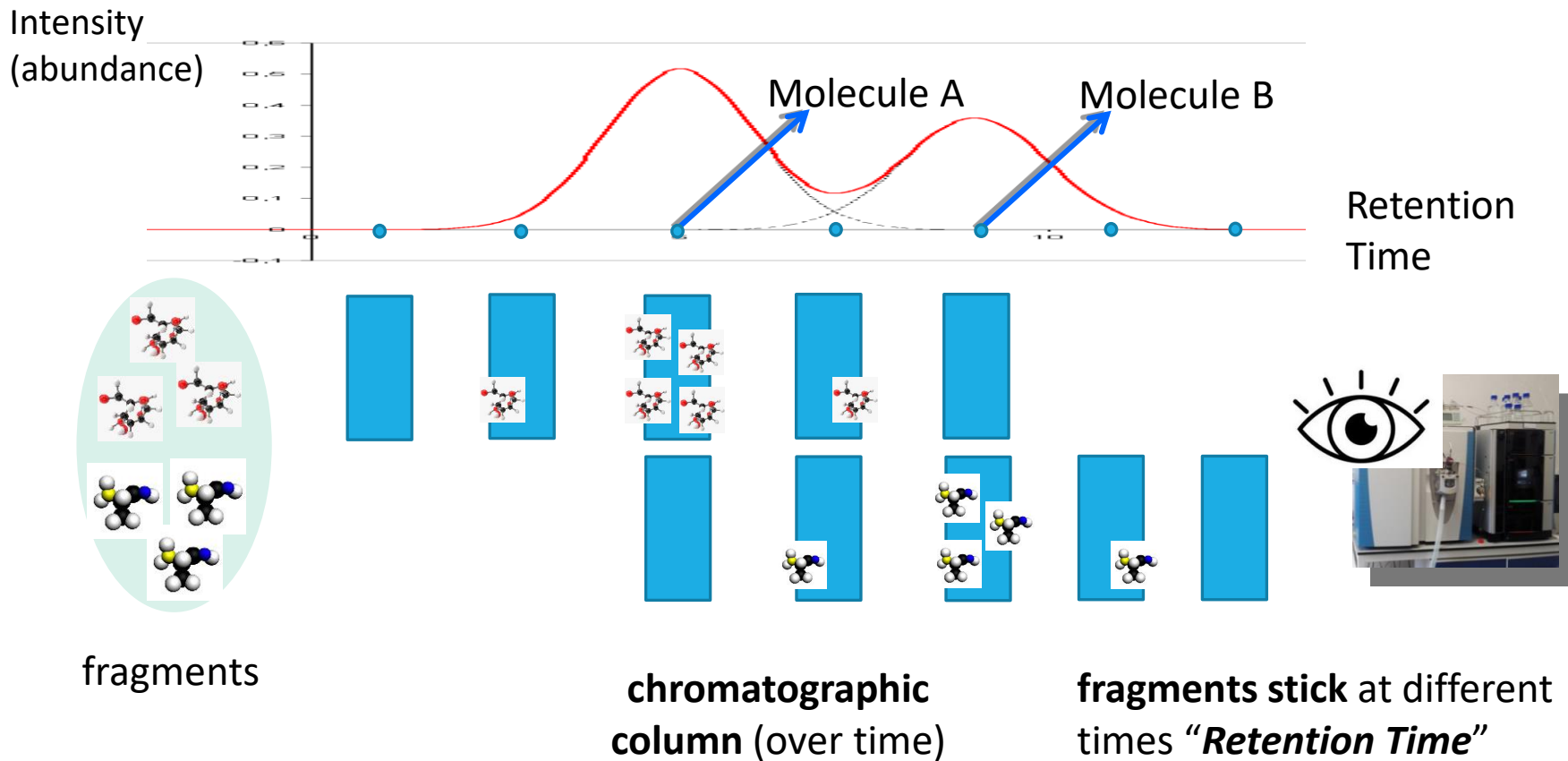
UniPR - Our research

- **WebApp Tool** : metabolite similarity matching with **Modified Cosine Similarity**
- ML Metabolite identification:
 - **Spec2Vec NN** (Word2Vec NN)
 - molecular embeddings from mass spectrometry data
 - trained model for molecular similarity matching
- ML Bioactivity prediction:
 - **UniRep (RNN)** → protein embeddings
 - **XGBoost classification**:
 - combined: molecular embeddings (spec2vec) + protein embeddings (Unirep)
 - metabolite + protein interactions = Active/Inactive (Papyrus)

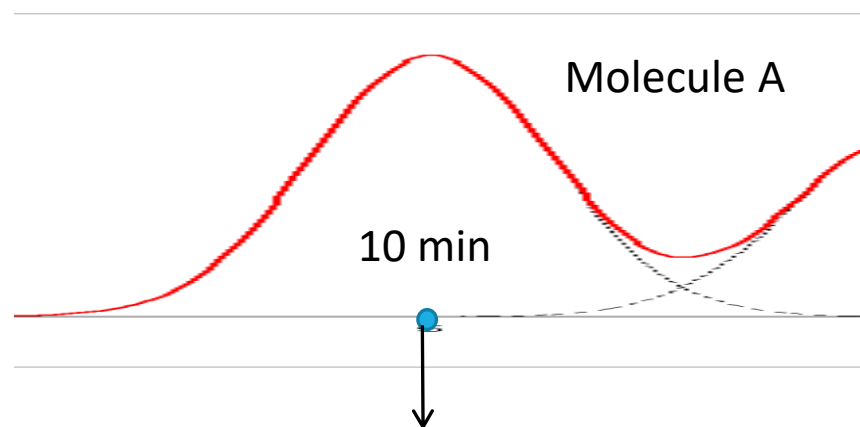
Data: Liquid Chromatography – Tandem Mass Spectrometry (LC-MS/MS)



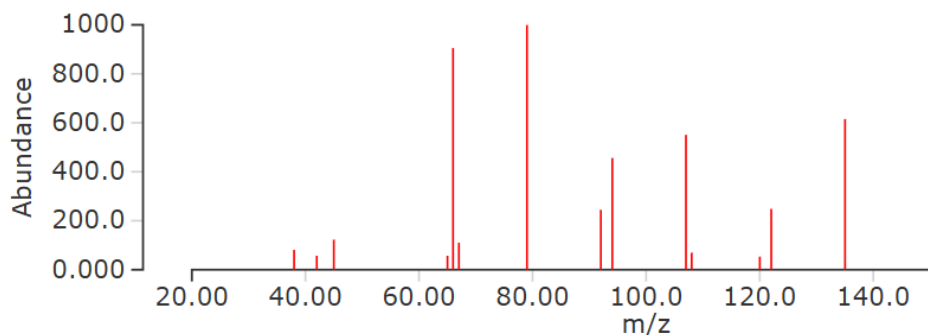
Chromatogram – molecules separation



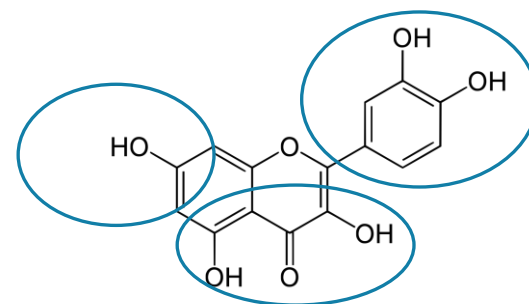
Mass Spectrum – molecule fingerprint



Molecule A – **Mass Spectrum** (e.g. retention time = 10 min)



small fragments $\longrightarrow$ big fragments



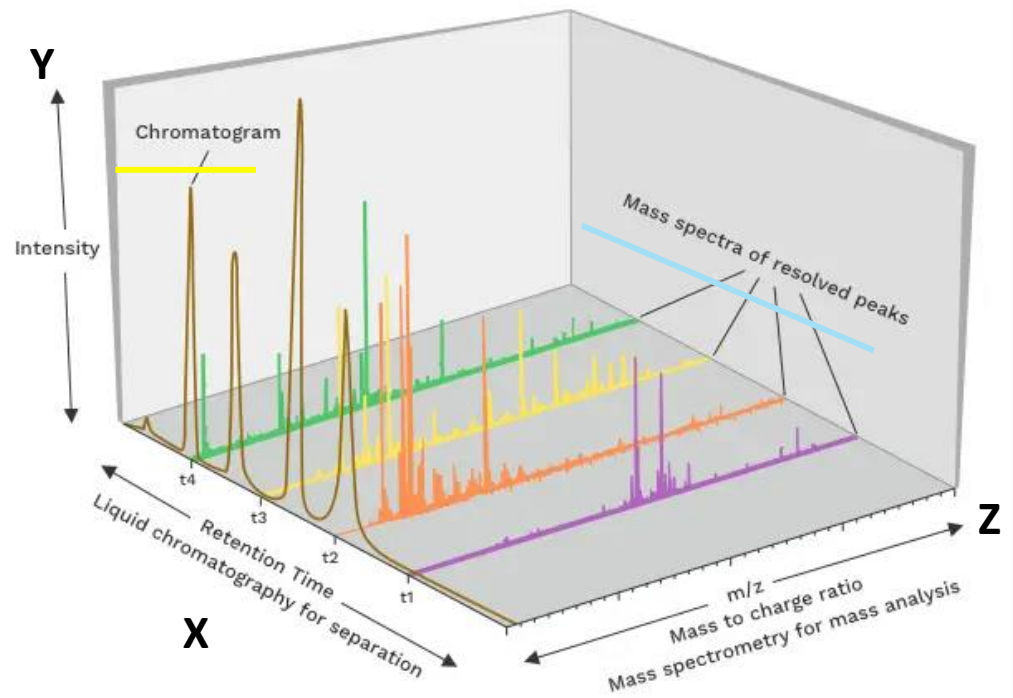
Data: Liquid Chromatography – Tandem Mass Spectrometry (LC-MS/MS)

Chromatogram (x,y)

- Separates different molecules
- **X axis:** “retention” time
- **Y axis:** amount of molecules

Mass Spectra (z,y)

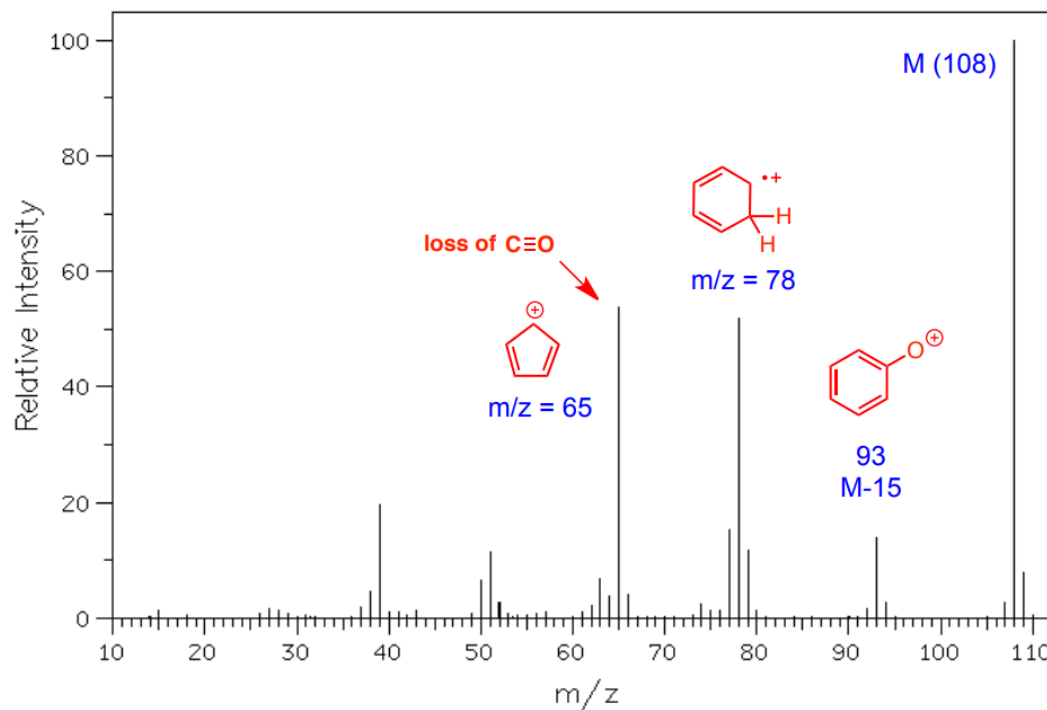
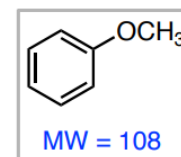
- Separates a molecule fragments
- **Z axis:** molecule fragment weight (**mass to charge ratio “m/z”**)
- **Y axis:** amount of a fragment



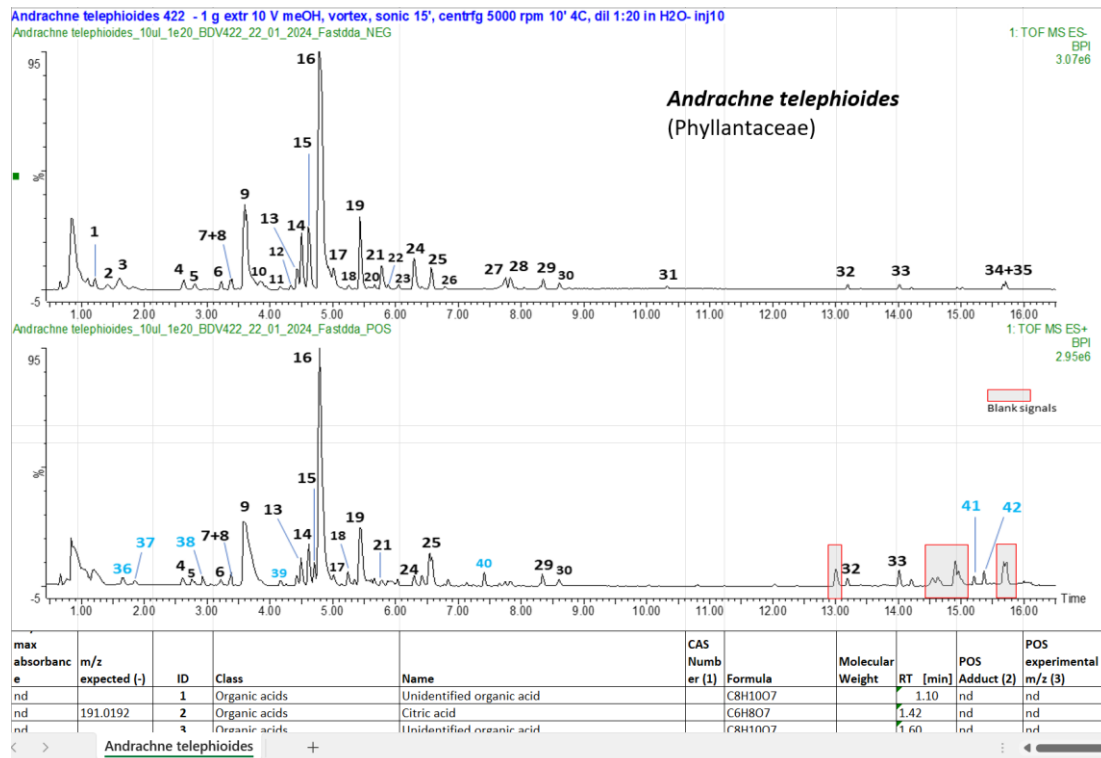
Metabolite identification - Experts annotation (manual)

Aryl Ethers

anisole



Metabolite identification - Experts annotation (manual)



Positive
ionization

Negative
ionization
(added in
blue)

Identification by
matching with a
reference spectrum
"standard library"

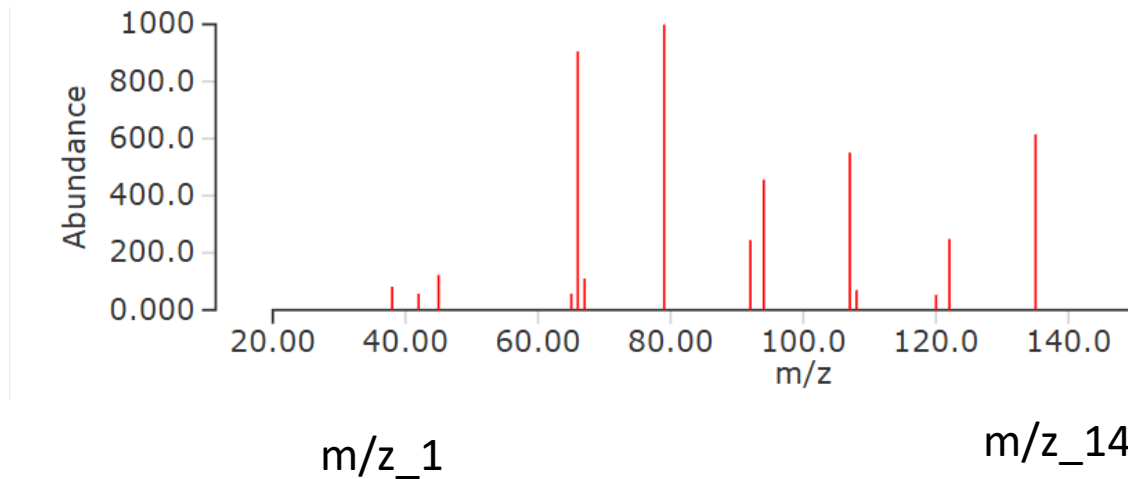
Molecular class

Name
(approximate)

Formula

Retention time

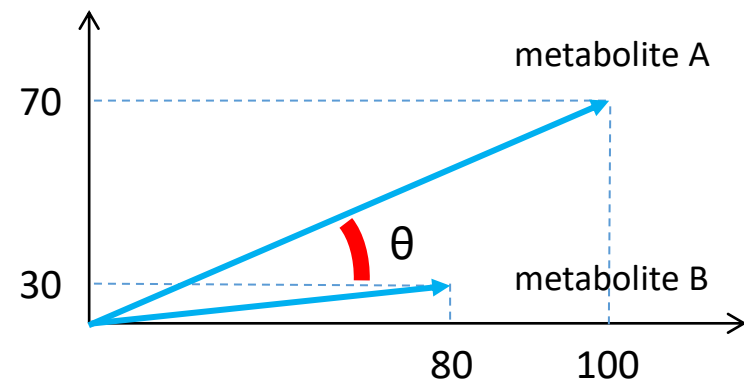
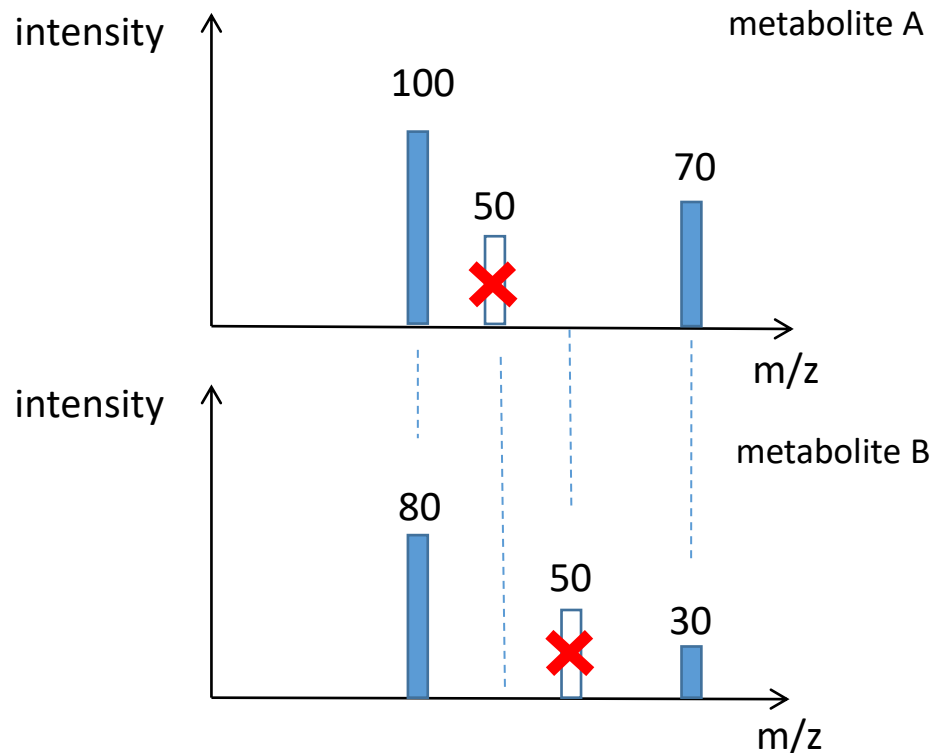
Metabolite identification - computational



- Mass Spectrum (MS) can be **used as a vectors** [(m/z , peak)]
 - m/z : [38.0 , 42.5, 49.2, ... , 135.0]
 - **peak intensity** : [100.0, 99.5, 425.3, ..., 602.0]

Metabolite identification - computational

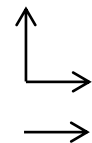
- Metabolite Matching : **Cosine Similarity**



cosine similarity

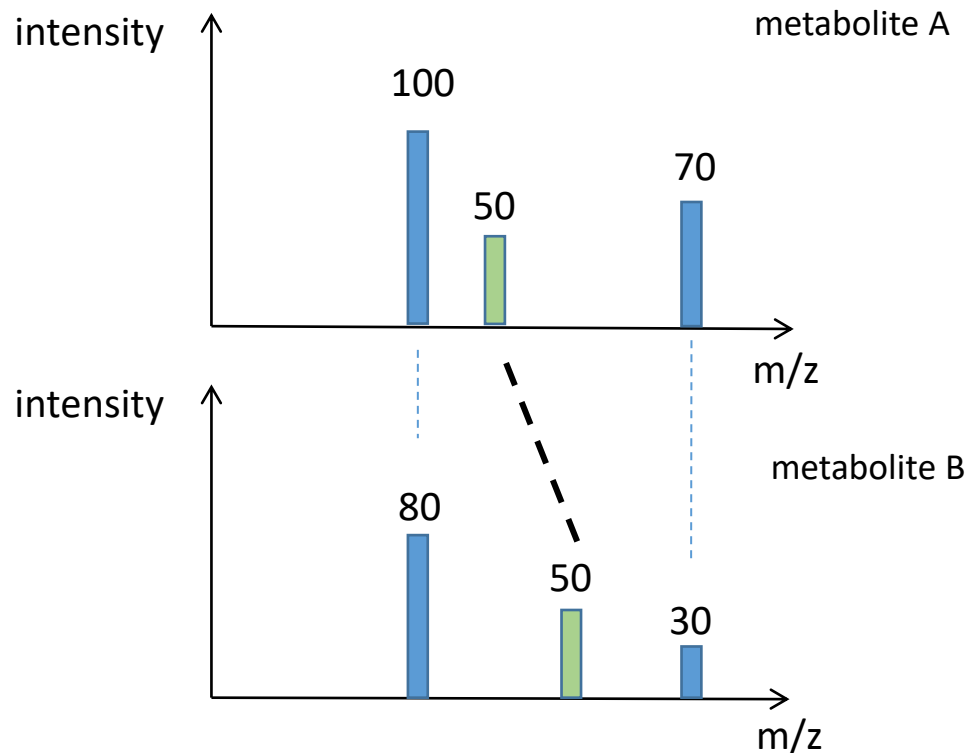
$$\cos(\theta = 90^\circ) = 0 = 0\%$$

$$\cos(\theta = 0^\circ) = 1 = 100\%$$



Metabolite identification - computational

- Metabolite Matching : **Modified Cosine Similarity**



Adds a **tolerance level**:

- + more matches
- + no training (formula)

- more False Positives

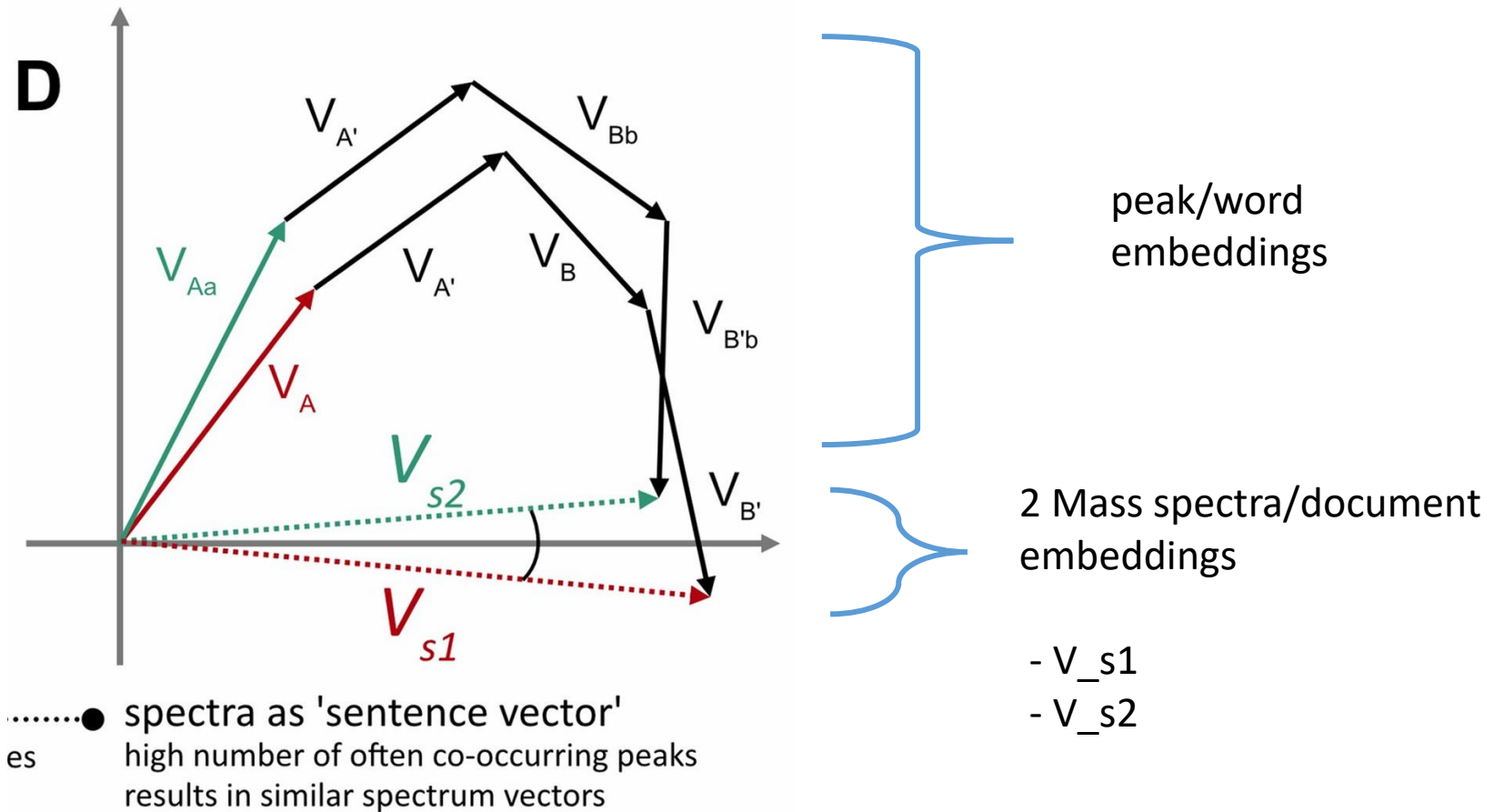
B4Web Tool:

- Database search

Metabolite identification - Spec2Vec NN

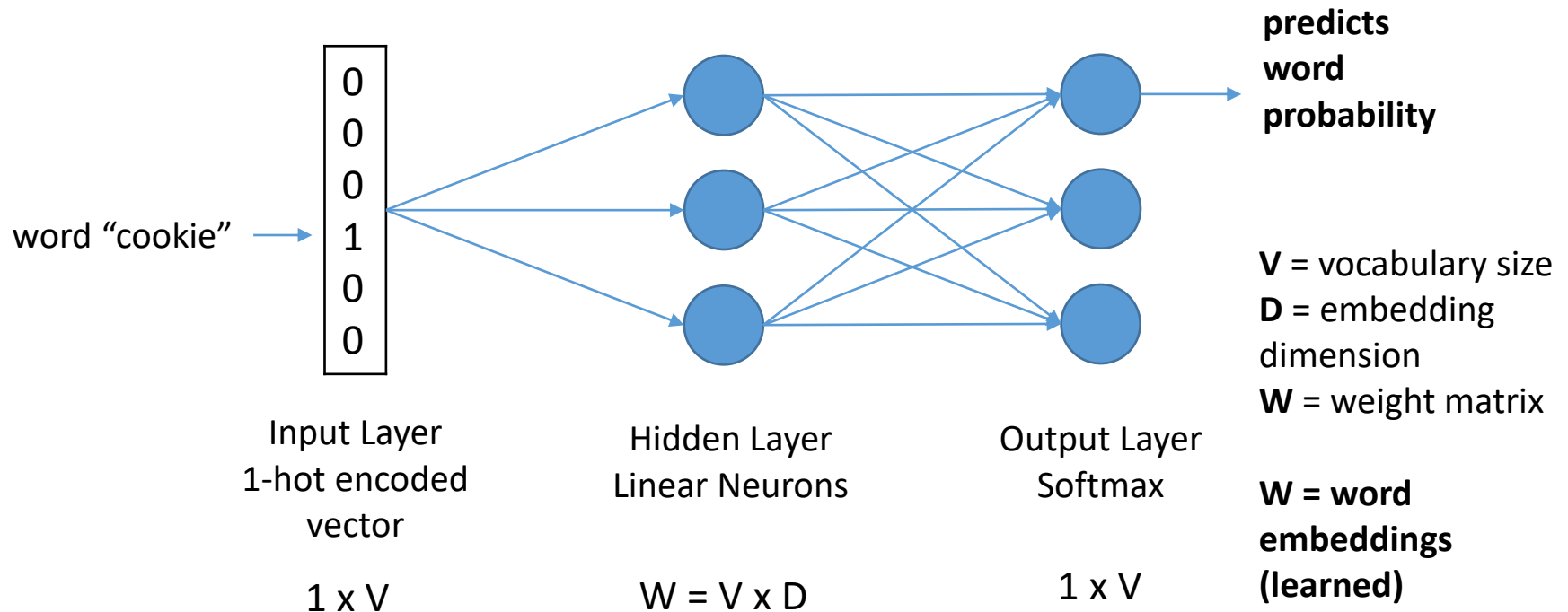
- **Adapts** Word2Vec NN to mass spectrometry
- **Each peak** in a mass spectrum is converted into a “**word**”
- **Each spectrum** becomes a “**document**” of “words”
- Using Word2Vec approach **learns an embedding for each word** based on its **context**
 - peaks co-occurrence
- Mass Spectra with similar structure have similar co-occurrence of peaks/words and **similar embeddings**

Metabolite identification - Spec2Vec NN



Word2Vec - Quick Reminder

- Word2Vec is a shallow, 2-Layer Neural Network



Word2Vec - Quick Reminder

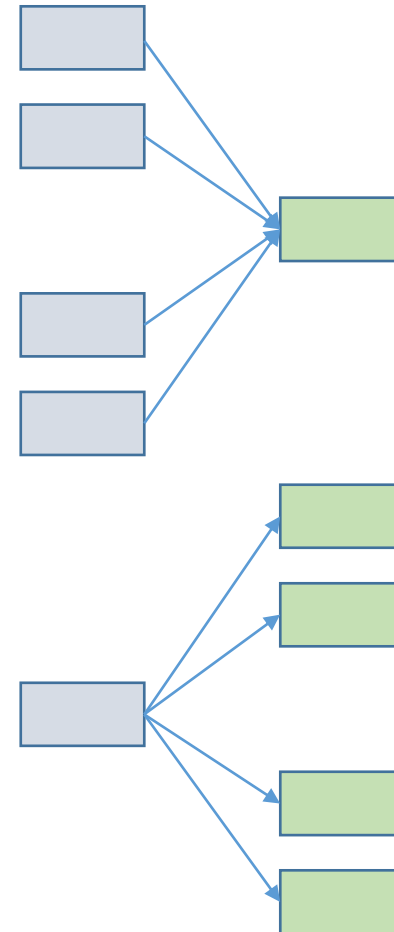
Why Word2Vec Learns Similar Embeddings

- Words that appear in **similar contexts** should produce **similar predictions**.
- To make their predictions similar, the model learns **similar hidden-layer vectors** (W matrix).
- learns similar embeddings for words used in similar ways

Word2Vec - Quick Reminder

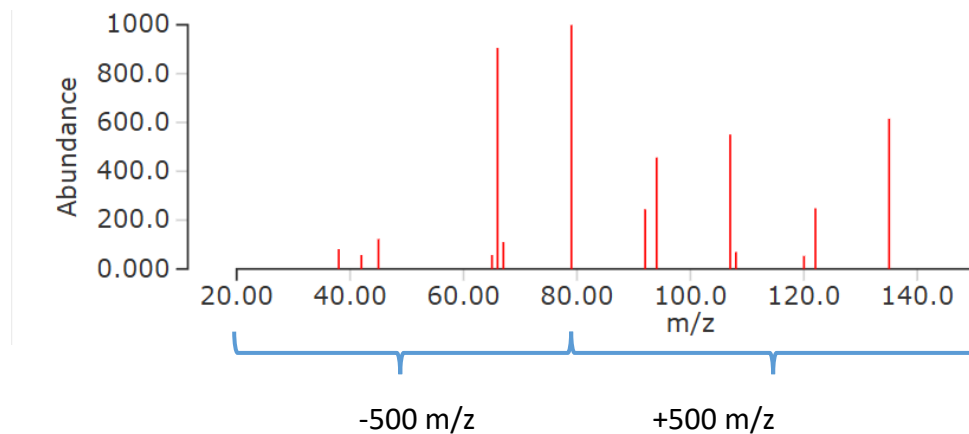
Two Word2Vec training strategies

- **CBOW** (Continuous Bag of Words):
 - Input: surrounding context words
 - Predicts the missing center word
- **Skip-gram**:
 - Input: center word
 - Predicts the surrounding context words



Spec2Vec - CBOW & Skip-gram

- Context is defined by a **m/z window**
 - usually **+/- 500 m/z** → includes all peaks
 - spec2vec allows to change the window size



Our Spec2Vec model

- **Dataset** obtained by intersection of:
 - **COCONUT**: database of **natural products**
 - List of metabolites
 - Download: <https://coconut.naturalproducts.net/download>
 - **MassBankEurope**: database of **LC-MS/MS spectral data**
 - Github data: <https://github.com/MassBank/MassBank-data>



Our Spec2Vec model

- **Training data:**



- Dataset (COCONUT+MassBankEU)
- **GNPS2** spectral libraries : <https://external.gnps2.org/gnpslibrary>
- **ALL_GNPS.json** → all MS/MS Spectra: **2,650,005**

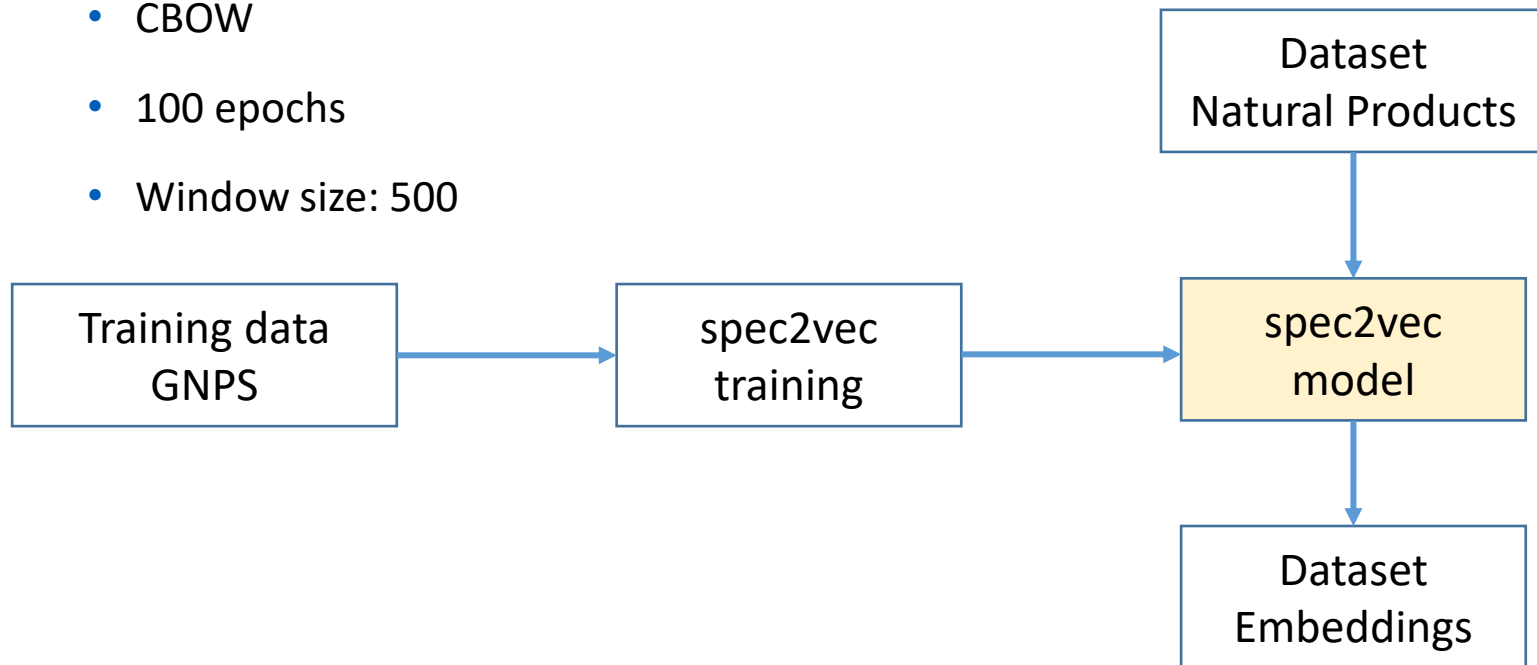
Our Spec2Vec model

- **Preprocessing:**
 - **matchms** → python library for MS data
 - following **spec2vec supplementary material**
 - Peak intensity to relative intensity 0 - 100%
 - Removed low intensity peaks (noise)
 - Removed spectra with < 10 peaks
- **Split** by ionization energy (positive/negative)
 - Models should be used and trained with data of the **same ionization energy**

Our Spec2Vec model

- **Spec2Vec training parameters:**

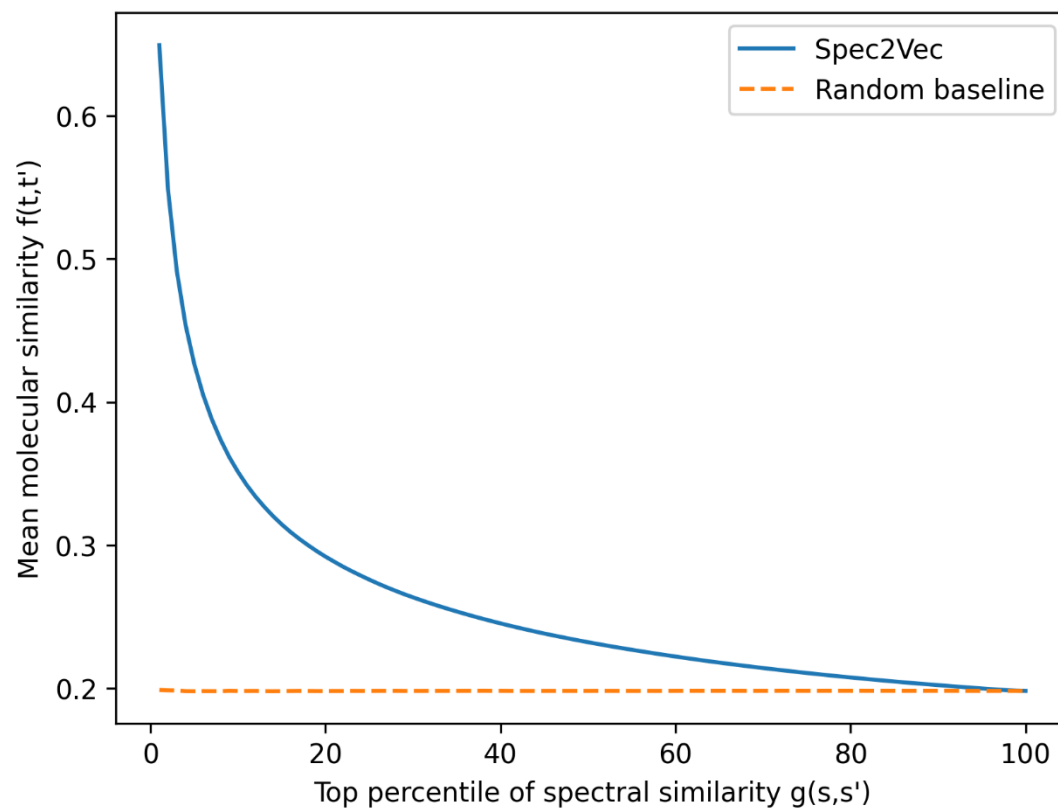
- Embedding size: 300 (default)
- CBOW
- 100 epochs
- Window size: 500



Our Spec2Vec model

Control for **biological
relevance:**

- **Spectral similarity vs
Structural Similarity**
- Cosine similarity
embeddings vs SMILES
fingerprint



Our Spec2Vec model

Control for **biological relevance**:

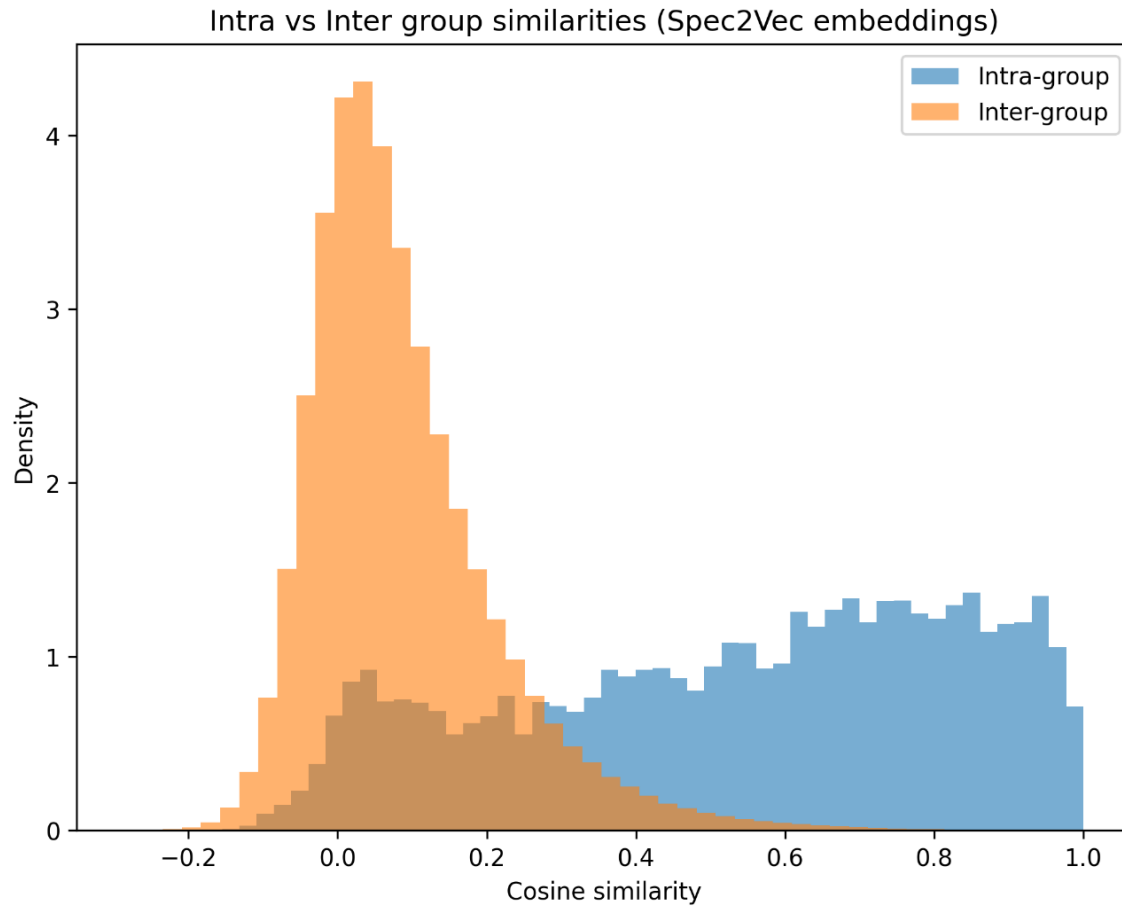
- **Intra-similarity**

Average Pairwise cosine similarity between embeddings of molecules with the same backbone (label)

- **Inter-similarity**

Average Pairwise cosine similarity between embeddings of molecules with different backbone (label)

Our Spec2Vec model



Our Spec2Vec model

Test for different embedding sizes

Embedding Size	Mean Intra Similarity	Mean Inter Similarity	AUC
300	0.5802	0.1607	0.8758
400	0.5523	0.1125	0.8889
500	0.5378	0.0862	0.8988
600	0.532	0.0731	0.9031
700	0.5276	0.0673	0.9044
1000	0.5259	0.0638	0.9052

Bioactivity identification

Vague definition of Bioactivity

- It refers to **any “detected” biological effect** of a metabolite
- Tested with experiments called “Bioassay”
 - **Untargeted:**
 - e.g. a metabolite makes cells mutate faster
 - qualitative measures: “+++”, “++”, “-” (not a real unit measure)
 - **Targeted** (our choice):
 - e.g. **metabolite enhances or inhibits a target protein**
 - quantitative measure : IC50, EC50
 - IC50: molecule concentration to inhibit 50% of the targets (opposite to EC50)

Bioactivity identification - Papyrus

- Article : “**Papyrus: a large-scale curated dataset aimed at bioactivity predictions**” 2023 - Béquignon & al. - Journal of Cheminformatics
- **Collect bioactivities** from public databases:
 - 60 million **metabolite - protein interactions**
- Creates a collection of high-confidence bioactivities
 - called Papyrus ++

Bioactivity identification - Papyrus

- Train a Deep Neural Network to predict bioactivities:
 - **Regression DNN:** (metabolite + protein) = **pChEMBL**
 - **Classification DNN:** (metabolite + protein) = **Active/Inactive**
- pChEMBL $\rightarrow$ bioactivity value (continuous)
- (metabolite + protein) is a **concatenation of descriptors**
 - **Metabolite fingerprint** from known SMILES molecular structure (python RDKit)
 - **Protein embedding** from UniRep (Recurrent Neural Network)

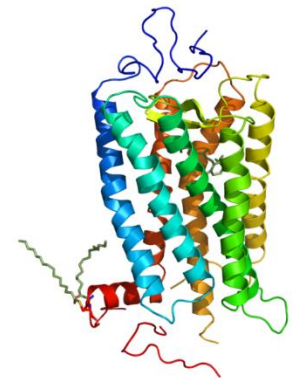
Bioactivity identification

Our method:

- Since Papyrus predicts bioactivities **for known molecules**
- We decided to predict bioactivities **for unknown natural products**
- By **creating a descriptor** using a concatenation of
 - molecule embeddings (Spec2Vec)
 - protein embeddings (UniRep)
- Bioactivities: Papyrus++
- Classification using XGBoost

UniRep - protein embeddings

- **Learns protein embeddings from protein sequences** of amino-acids
- The **meaning** of each new amino-acid **depends on the past context**
- Biological motivation:
 - **protein function depends on their 3D structure**
 - **3D structures depend on the sequence** of amino-acids
 - Example: **alpha-helix**, loops, protein domains
- **Recurrent Neural Network** with multiplicative Long Short Term Memory (mLSTM)



UniRep - protein embeddings

- **Recurrent Neural Network:**
 - Input: amino-acid from a protein sequence
 - Processes **1 amino-acid each iteration**
 - **1-hot encoding** vector (**1x20 amino-acids**)
 - Has **memory** to remember past context
- Long Short Term Memory (LSTM)
 - cell state (long-term memory) c_t
 - hidden state (short-term) h_t
 - UniRep states dimension is a **1x1900 vector**

UniRep - protein embeddings

multiplicative LSTM:

- multiplicative modulation term

$$m_t = (W_{mx}x_t * W_{mh}h_{t-1})$$

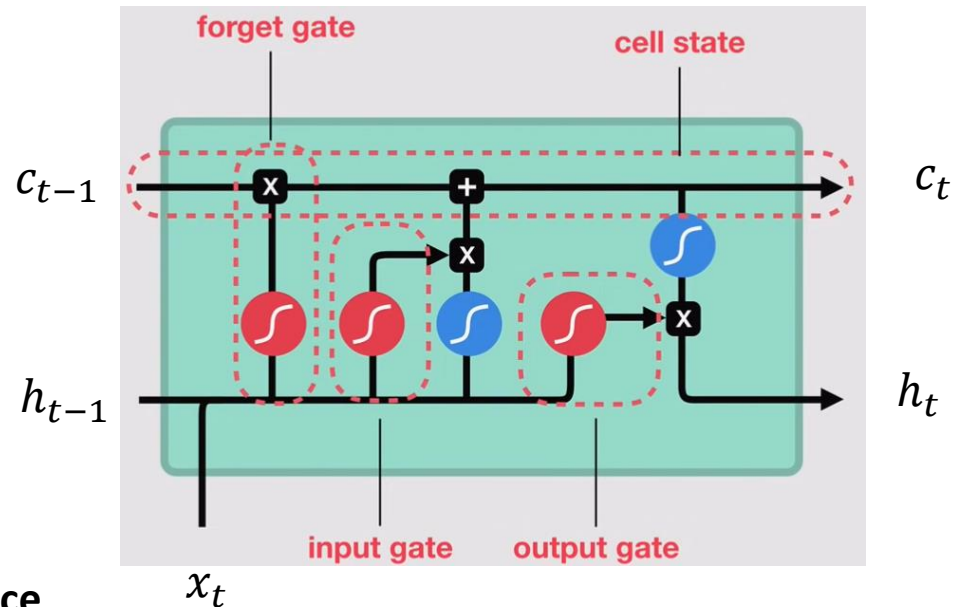
- input influence * context influence

- Classic LSTM:

- input influence + context influence

- Applies to **all the gates**

- m_t instead of h_{t-1}



$$g_t = \tanh(W_g x_t + U_g m_t + b_g)$$

$$c_t = f_t * c_{t-1} + i_t g_t$$

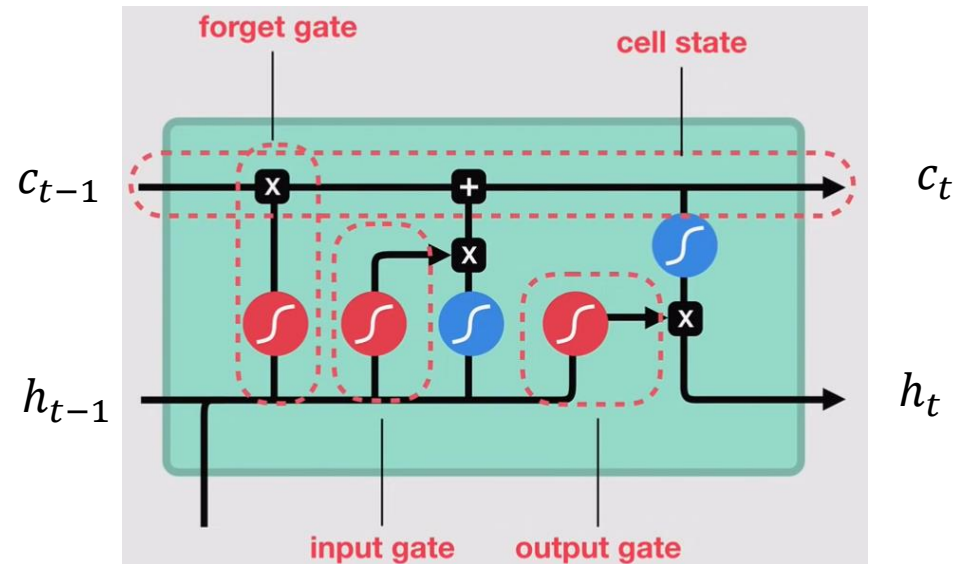
$$h_t = o_t * \tanh(c_t)$$

UniRep - protein embeddings

multiplicative LSTM:

- Forget gate:
- how much past context to keep

$$f_t = \sigma(W_f x_t + U_f \mathbf{m}_t + b_f)$$

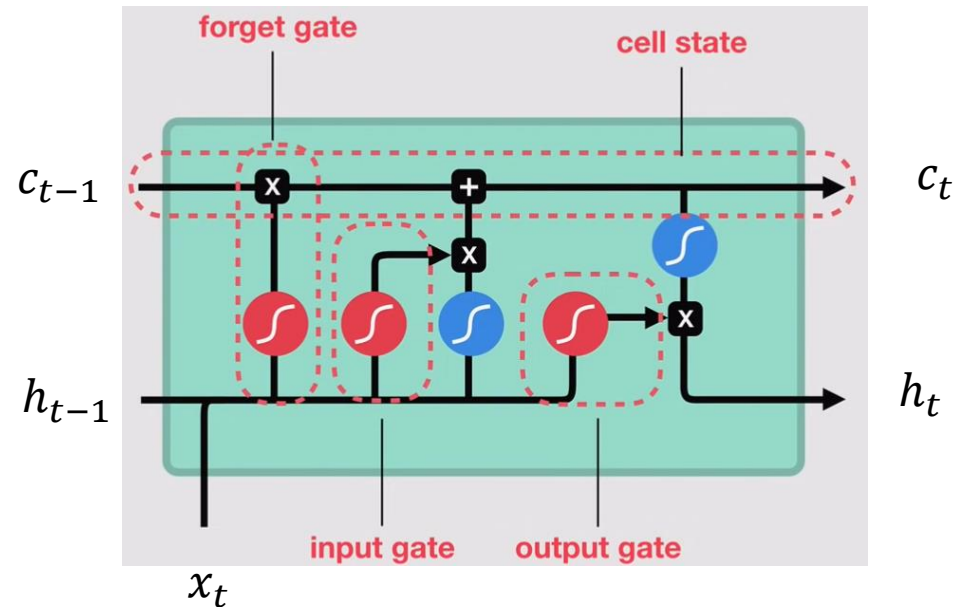


UniRep - protein embeddings

multiplicative LSTM:

- Input gate:
- new information to add

$$i_t = \sigma(W_i x_t + U_i \mathbf{m}_t + b_i)$$



$$g_t = \tanh(W_g x_t + U_g \mathbf{m}_t + b_g)$$

$$c_t = f_t * c_{t-1} + i_t g_t$$

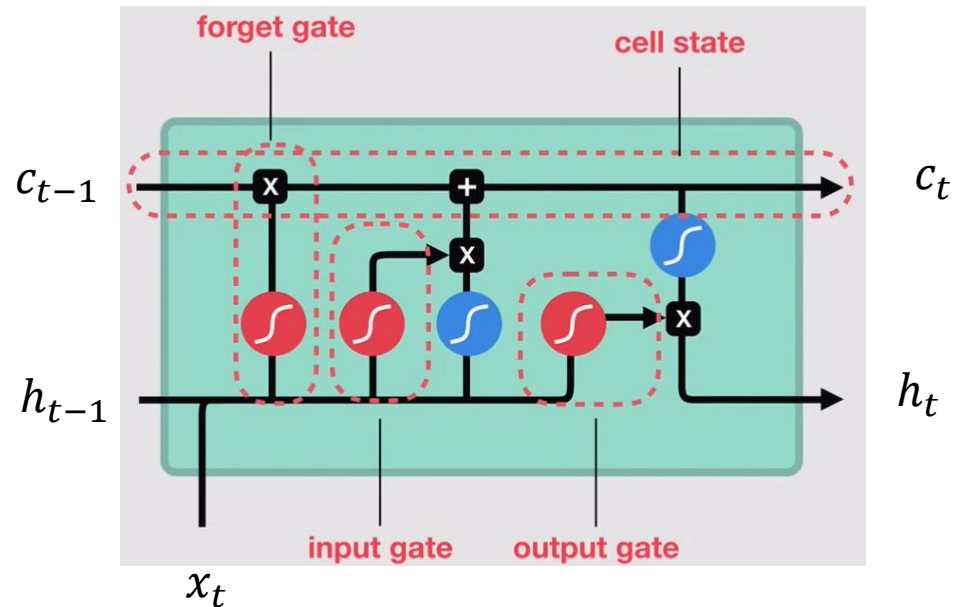
$$h_t = o_t * \tanh(c_t)$$

UniRep - protein embeddings

multiplicative LSTM:

- Output gate:
- what part of the memory to output

$$o_t = \sigma(W_o x_t + U_o \mathbf{m}_t + b_o)$$



$$g_t = \tanh(W_g x_t + U_g \mathbf{m}_t + b_g)$$

$$c_t = f_t * c_{t-1} + i_t g_t$$

$$h_t = o_t * \tanh(c_t)$$

UniRep - protein embeddings

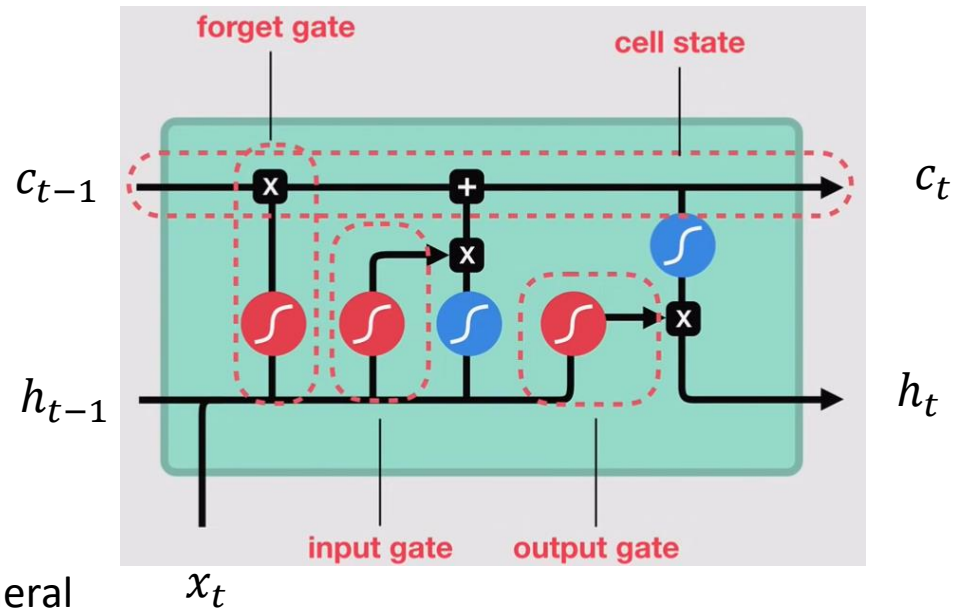
Output:

- final hidden state h_t
- learned protein representation

UniRep saves all hidden states:

- **Averages hidden states** to capture general representation of a protein

$$\bar{h} = \frac{1}{T} \sum h_t$$



$$g_t = \tanh(W_g x_t + U_g \mathbf{m}_t + b_g)$$

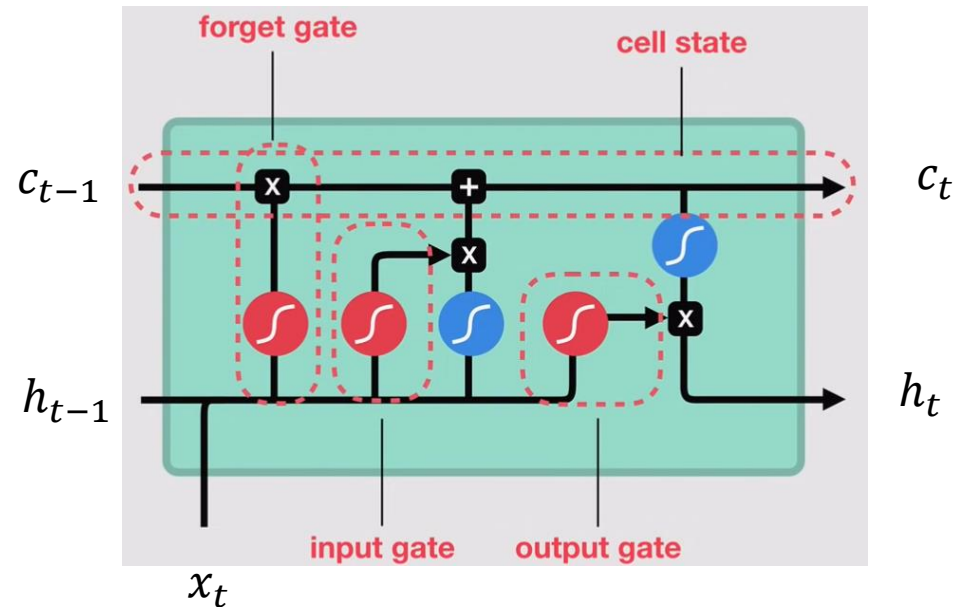
$$c_t = f_t * c_{t-1} + i_t g_t$$

$$h_t = o_t * \tanh(c_t)$$

UniRep - protein embeddings

Complete UniRep embedding:

- Vector $1900 \times 3 = 5700$
- 1900 average of hidden states
- 1900 standard deviation
- 1900 final hidden state



$$g_t = \tanh(W_g x_t + U_g \mathbf{m}_t + b_g)$$

$$c_t = f_t * c_{t-1} + i_t g_t$$

$$h_t = o_t * \tanh(c_t)$$

Bioactivity classification

- Our descriptor
 - metabolite embedding → 500 dim
 - protein embedding → 5700 dim
 - concatenation: 6200 dim
- Challenges:
 - Unbalanced features: 500 vs 5700
 - Different feature scale (spec2vec vs UniRep)
 - embeddings have some redundancy → adds noise to classification

Bioactivity classification

- Processing descriptors
 - Feature scaling : standardization (mean:0, std:1)
 - PCA for each embedding, same dimensionality
- Classification with XGBoost → negative ionization dataset

PCA	MCC	Precision	Recall	Accuracy	F1 score
500	0.9189	0.9695	0.9262	0.9625	0.9473
256	0.9189	0.9694	0.9264	0.9625	0.9473
128	0.9233	0.9709	0.9306	0.9645	0.9503
64	0.9255	0.9751	0.9291	0.9655	0.9515
32	0.9207	0.9710	0.9270	0.9633	0.9484

Bioactivity classification

- **In our case** we prioritize the Recall
 - Higher Recall → **more Active bioactivities predicted**
 - Cons: a few False Positives → less important: waste a bit of time
 - For a laboratory **missing** to test a good bioactivity **is worse** than wasting an experiment on a False Positive
 - **Choice : PCA = 128**

PCA	MCC	Precision	Recall	Accuracy	F1 score
500	0.9189	0.9695	0.9262	0.9625	0.9473
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64	0.9255	0.9751	0.9291	0.9655	0.9515
32	0.9207	0.9710	0.9270	0.9633	0.9484

Take Home

- **Molecular data are challenging to analyze**
 - Metabolite identification → similarity
 - Bioactivity identification → classification
- **Neural Networks can be leveraged to learn from context**
 - spec2vec (word2vec) → molecular embeddings → co-occurrence context
 - UniRep (RNN) → protein embeddings → past sequence context
- **Combining embedding representations in a descriptor** can be used to train further Machine Learning models
 - XGBoost Classifier
- Laboratory Experts **may prefer high Recall models** despite overall lower performances

